

# **IEORE4004**

# **Optimization Models and Methods**

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# Outline

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- About Project 1
- Introduction to Integer Programming
  - Comparison of IPs to LPs
- Common Integer Programming Problems
  - Knapsack Problem
    - Multiple Knapsack
    - Multiple Multi Dimensional Knapsack
- Modeling Logical Constraints

# Project 1: Eliminating Child Care Deserts in New York City through Optimization Modeling

**Childcare deserts:** regions where the demand for licensed childcare far exceeds available slots.

This classification is determined by several factors, including:

- The total number of available childcare slots;
- The population of children requiring care;
- The percentage of employed parents;
- The average income in the area.



# Project 1: Eliminating Child Care Deserts in New York City through Optimization Modeling

## Problem 1: Budgeting

NYC aims to eliminate childcare deserts by increasing the number of childcare slots across all regions. To achieve this, NYC will provide funding to either:

1. Build new facilities, or
2. Expand existing ones

There are three possible sizes for new construction, each with approximate capacities and costs:

| Facility Size      | Max # of Slots (Ages 0-5) | Cost of New Facility (\$) to NYC |
|--------------------|---------------------------|----------------------------------|
| 100 slots (Small)  | 50 slots                  | 65,000                           |
| 200 slots (Medium) | 100 slots                 | 95,000                           |
| 400 slots (Large)  | 200 slots                 | 115,000                          |

# Project 1: Eliminating Child Care Deserts in New York City through Optimization Modeling

## Problem 1: Budgeting

In the first step, NYC considers an idealistic (optimistic) scenario that assumes:

- New facilities can be built anywhere in NYC. Any number of new facilities may be constructed in any area.
- Expansions are limited to a maximum of 120% of current capacity, or up to 500 slots per facility.
- Expansion costs depend on the scale of the increase. Larger facilities are cheaper to expand on a per-slot basis.
- To expand capacity by 100% or more, NYC must pay a baseline cost of \$20,000, plus \$200 per existing slot.
- New slots for children under age five require an extra \$100 per slot for specialized equipment.

# Project 1: Eliminating Child Care Deserts in New York City through Optimization Modeling

## Problem 2: Realistic Budgeting & Location

### Recommendations on costs

- due to space limitations, it is not possible to expand the capacity at existing facilities beyond 20%,
- the larger the expansion, the higher the marginal cost per scale of the expansion.

### Recommendations on locations

- Within each area, no two facilities, whether new or existing, can be located too close to one another. The minimum distance between any facility should be at least 0.06 miles

# Project 1: Eliminating Child Care Deserts in New York City through Optimization Modeling

## Important Dates

- **February 27, Friday:** Initiate contact with your team



- **March 9, Monday:** In-Class Project Work Time
- **March 16, Monday:** IEOR4004 Team Check in
- **March 25, Wednesday:** Final Report & Team Assessment

# **Introduction to Integer Programming**



# Integer Programming

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- Many practical optimization problems involve discrete decisions.
- A natural framework for modeling discrete decisions is integer programming.
- Integer programs can have a mixture of continuous, integer, and binary (0 or 1) variables.

# Linear vs Integer Programming

An integer program (IP) is a type of mathematical optimization problem in which *some or all* the decision variables are required to take integer values

## Standard Form

Linear Program (LP):

$$\max c^T x$$

$$\begin{array}{l} Ax \leq b \\ x \in \mathbb{R}^n \end{array}$$

Integer Program (IP):

$$\max c^T x$$

$$\begin{array}{l} Ax \leq b \\ x \in \mathbb{Z}^n \end{array}$$

# Types Of Integer Programs

- Pure Integer Program (PIP or IP)
- Binary Integer Program (BIP) or 0-1 Program
- Mixed-Integer Program

$$\max c^T x$$

$$\begin{array}{ccc} Ax & \leq & b \\ x & \in & \mathbb{Z}^n \end{array}$$

$$\max c^T x$$

$$\begin{array}{ccc} Ax & \leq & b \\ x & \in & \{0, 1\}^n \end{array}$$

$$\max \quad c^T x \quad + d^T y$$

$$\begin{array}{ccc} Ax & + By & \leq b \\ x & & \in \mathbb{Z}^n \\ & y & \in \mathbb{R}^n \end{array}$$

# Solving LPs vs IPs

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Solving an IP model is generally more difficult than solving its LP counterpart, primarily because:

## **1. The feasible solution space for IPs can be very large because it consists of discrete points.**

- As the number of integer variables or constraints increases, the number of potential combinations of integer values can grow exponentially. This is especially true for binary integer programs where the number of possible combinations is  $2^n$  for  $n$  binary variables.

# Solving LPs vs IPs

---

Solving an IP model is generally more difficult than solving its LP counterpart, primarily because:

**2. While there are efficient algorithms for solving LPs (like the Simplex method or interior point methods), there's no universally efficient algorithm for IPs.**

- Common techniques for solving IPs include branch and bound, branch and cut, and cutting plane methods. These methods can be computationally intensive, especially for large-scale problems.

# Solving LPs vs IPs

---

Solving an IP model is generally more difficult than solving its LP counterpart, primarily because:

## **3. The feasible region of IPs are not convex.**

- The feasible region defined by LP constraints is convex, which means that if you take any two feasible solutions and form a weighted combination of them, that combination is also feasible.



# Solving LPs vs IPs

---

Solving an IP model is generally more difficult than solving its LP counterpart, primarily because:

## **4. In IPs, local optima (best solutions in a certain neighborhood) may not be the best overall solutions.**

- For LPs, any local optimum is also a global optimum. This means algorithms need to explore more of the solution space to ensure global optimality, making the process more time consuming.

# Linear vs Integer Programming

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Let's try to compare this LP to its IP counterpart:

LP

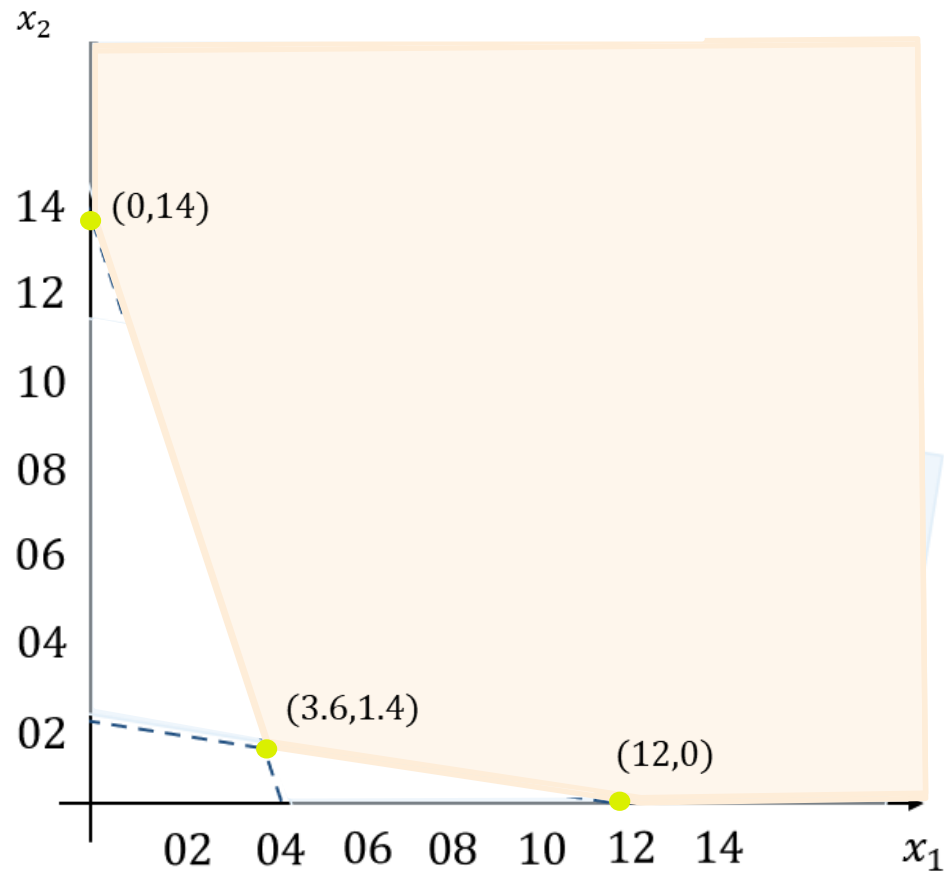
$$\begin{aligned} \min z &= 50x_1 + 100x_2 \\ \text{s.t.} \quad &7x_1 + 2x_2 \geq 28 \\ &2x_1 + 12x_2 \geq 24 \\ &x_1, x_2 \geq 0 \end{aligned}$$

IP

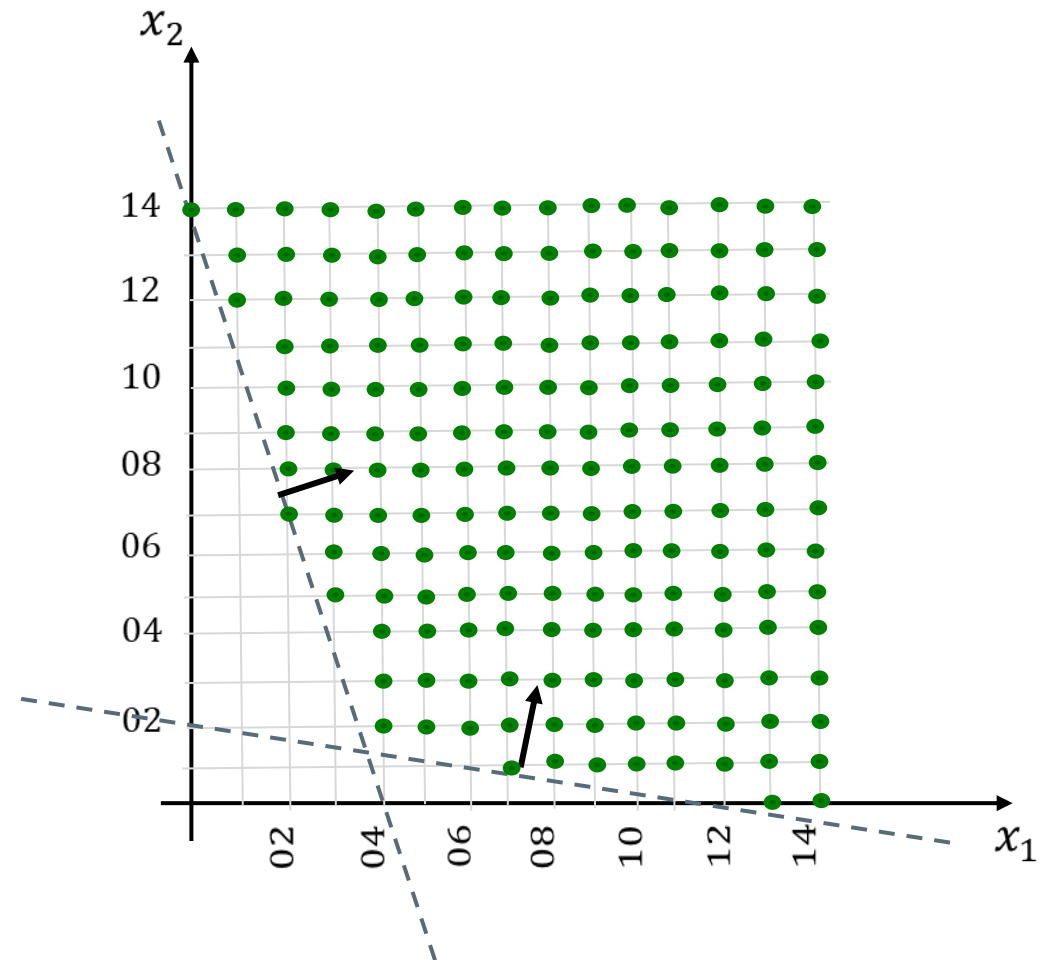
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# Linear vs Integer Programming

Feasible Region of the LP

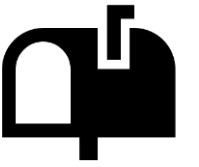


Feasible "Region" of the IP



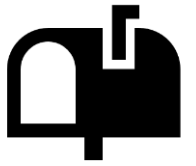
# A Scheduling Or Staffing Problem

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The **new post office** requires at least 17 employees on Mondays, 13 on Tuesdays, 15 on Wednesdays, 19 on Thursdays, 14 on Fridays, 16 on Saturdays, and 11 on Sundays. Union rules require employees to work 5 consecutive working days and 2 consecutive days off.

# A Scheduling Or Staffing Problem

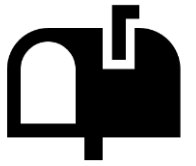


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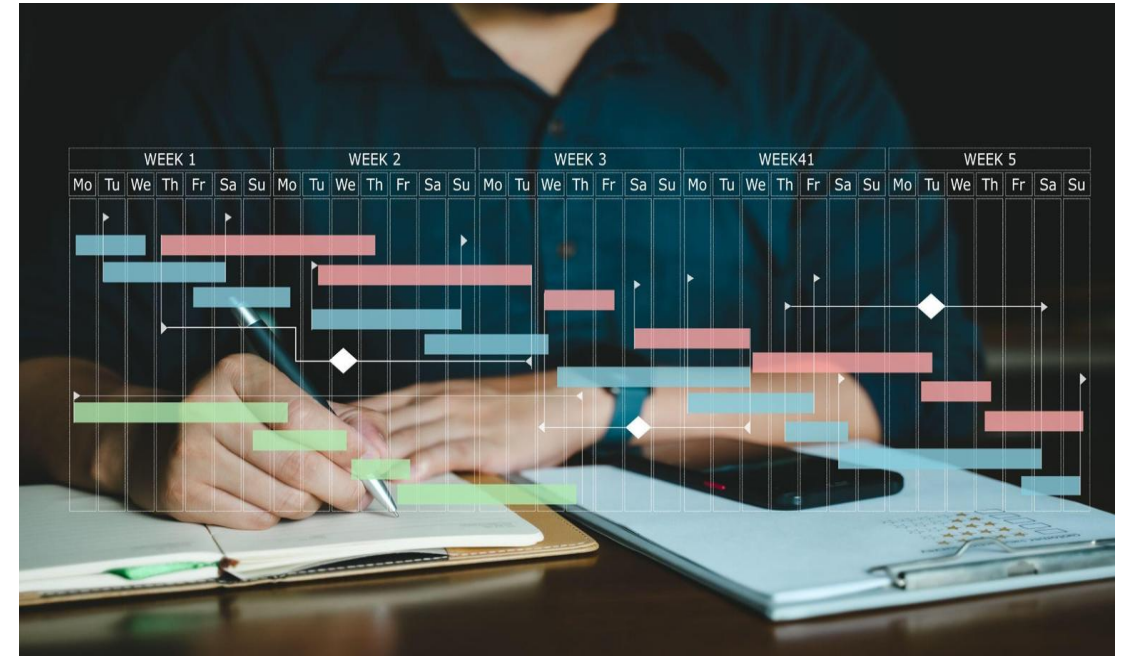
How many employees should you hire?

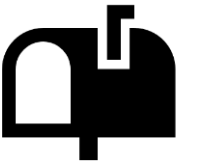
# Formulation: A Staffing Problem



Define the decision variables:

- $x_1$  is the number of workers who start on Monday and work until Friday,
- $x_2$  is the number of workers who start on Tuesday and work until Saturday,
- $x_3, x_4, \dots, x_7$  are defined similarly.



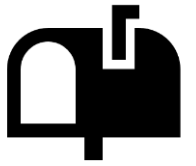


# Formulation: A Staffing Problem

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Objective Function  $\longrightarrow$   $\min \quad x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7$

# Formulation: A Staffing Problem



Objective Function  $\longrightarrow$   $\min x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7$

Constraints  $\longrightarrow$  s.t.

$$\begin{aligned} x_1 + x_4 + x_5 + x_6 + x_7 &\geq 17 \\ x_1 + x_2 + x_5 + x_6 + x_7 &\geq 13 \\ x_1 + x_2 + x_3 + x_6 + x_7 &\geq 15 \\ x_1 + x_2 + x_3 + x_4 + x_7 &\geq 19 \\ x_1 + x_2 + x_3 + x_4 + x_5 &\geq 14 \\ x_2 + x_3 + x_4 + x_5 + x_6 &\geq 16 \\ x_3 + x_4 + x_5 + x_6 + x_7 &\geq 11 \\ x_1, x_2, x_3, x_4, x_5, x_6, x_7 &\geq 0 \end{aligned}$$



# A Transportation Problem

---

An International Assistance Team wants to provide additional support and resources to an earthquake-affected area:

- There are two hubs with **100** and **200** specially trained and equipped volunteers waiting to be assigned to an affected area,
- The estimated demand for each affected area is **50, 100, 100** volunteers, respectively
- Estimated transportation cost per volunteer:

| Affected Areas: | 1     | 2     | 3     |
|-----------------|-------|-------|-------|
| Hub 1           | \$100 | \$120 | \$140 |
| Hub2            | \$110 | \$120 | \$130 |

# A Transportation Problem

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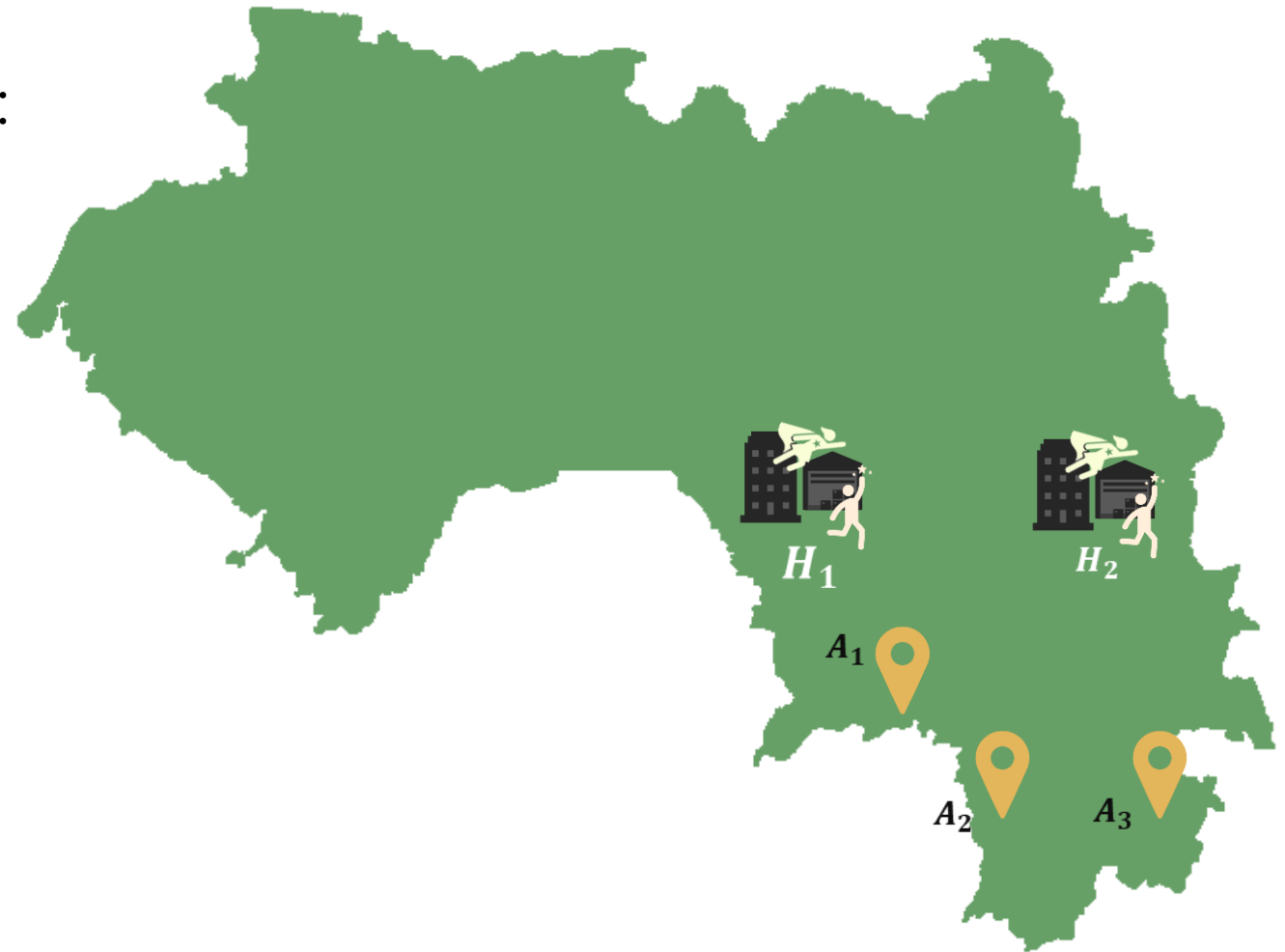
## Cost of:

Fuel, vehicles, aircraft, accommodation, meals, equipment, supplies, and training ...

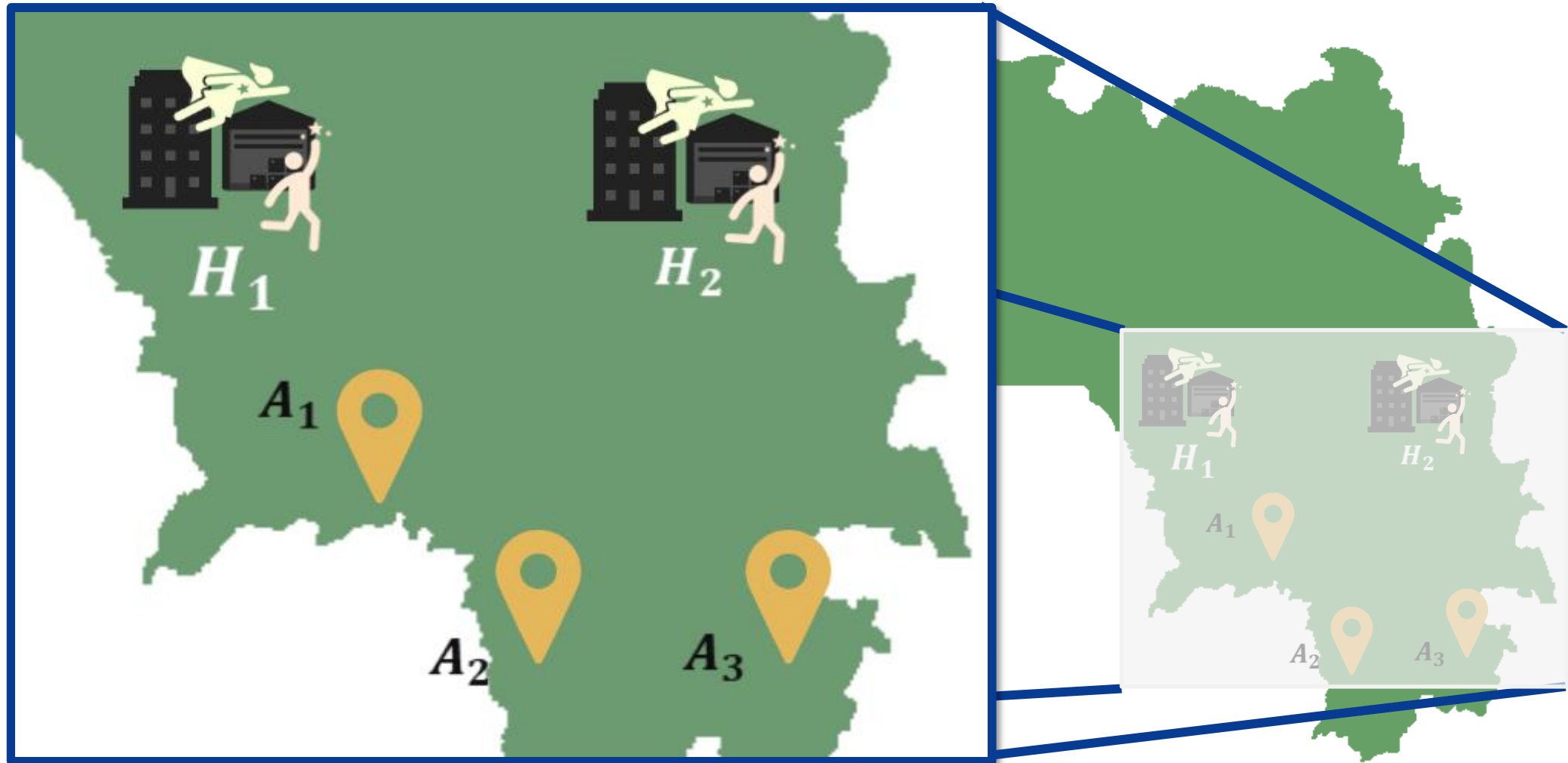
# How Can We *Minimize* The Total Transportation Cost While Satisfying The Demand?

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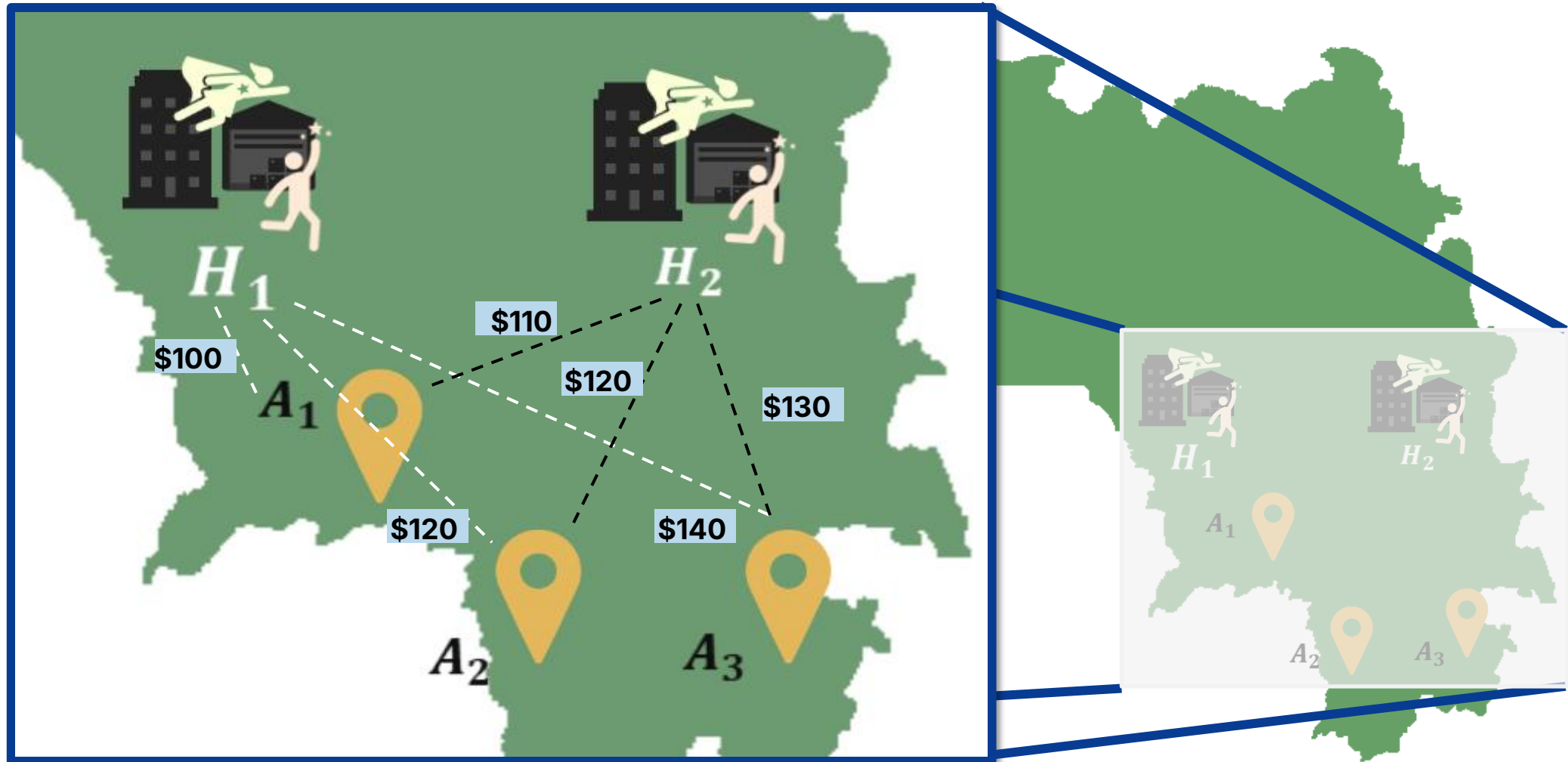
Let's take a look at the map:



# How Can We *Minimize* The Total Transportation Cost While Satisfying The Demand?



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# Formulation: A Transportation Problem

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## Decision variables:

$x_{i,j}$ : Number of volunteers to transport from hub  $i$  to earthquake-affected area  $j$

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## Objective:

$$\text{minimize } 100x_{1,1} + 120x_{1,2} + 140x_{1,3} + 110x_{2,1} + 120x_{2,2} + 130x_{2,3}$$

# Formulation: A Transportation Problem

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## Decision variables:

$x_{i,j}$ : Number of volunteers to transport from hub  $i$  to earthquake-affected area  $j$

## Objective:

$$\text{minimize} \quad 100x_{1,1} + 120x_{1,2} + 140x_{1,3} + 110x_{2,1} + 120x_{2,2} + 130x_{2,3}$$

## Constraints:

$$x_{1,1} + x_{1,2} + x_{1,3} \leq 100$$

$$x_{2,1} + x_{2,2} + x_{2,3} \leq 200$$

$$x_{1,1} + x_{2,1} \geq 50$$

$$x_{1,2} + x_{2,2} \geq 100$$

$$x_{1,3} + x_{2,3} \geq 100$$

$$\vec{x} \geq 0$$



# Formulation: A Transportation Problem

## Decision variables:

$x_{i,j}$ : Number of volunteers to transport from hub  $i$  to earthquake-affected area  $j$

## Objective:

$$\text{minimize } 100x_{1,1} + 120x_{1,2} + 140x_{1,3} + 110x_{2,1} + 120x_{2,2} + 130x_{2,3}$$

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$$x_{2,1} + x_{2,2} + x_{2,3} \leq 200$$

$$x_{1,1} + x_{2,1} \geq 50$$

$$x_{1,2} + x_{2,2} \geq 100$$

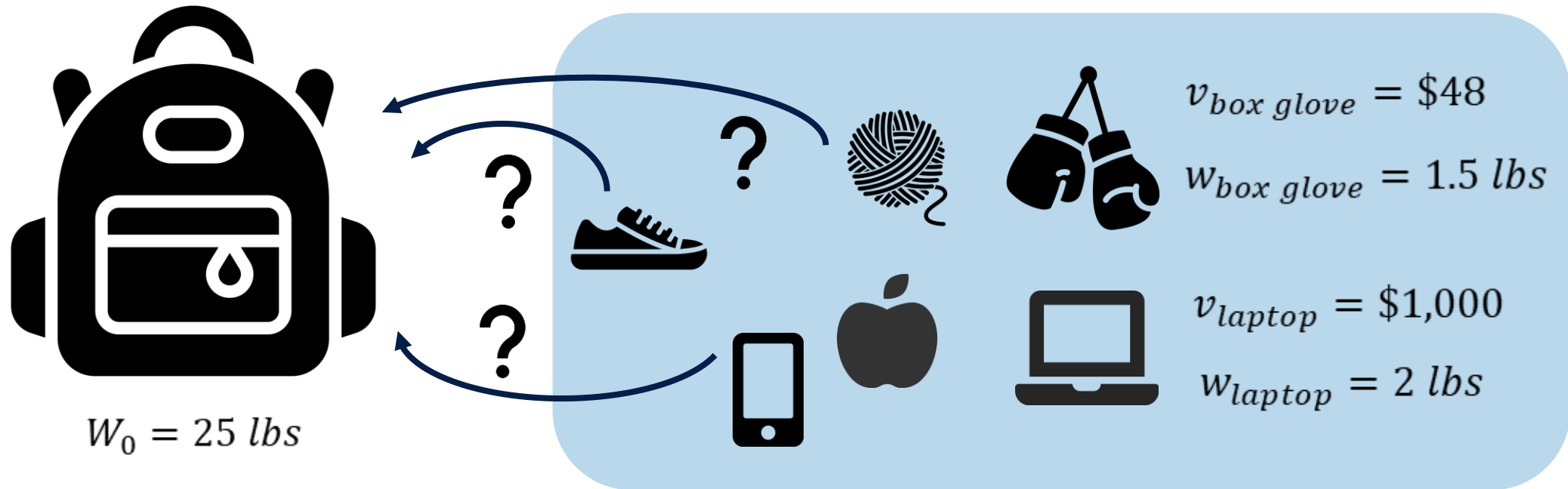
$$x_{1,3} + x_{2,3} \geq 100$$

$$\vec{x} \geq 0$$

**$\vec{x}$  can only take integer values  
(Cannot transport half a person!)**

*and  $\vec{x} \in \mathbb{Z}^+$*

# Knapsack Problem





# Knapsack Problem

## Given:

- objects  $1, \dots, n$ , where each object  $j$  has value  $v_j$  and weight  $w_j$ ;
- A knapsack (i.e., a bag) with capacity  $W_0$ .

## Find:

- Among all sets of objects, the ones which maximize the profit while considering the upper bound on weights ( $W_0$ )

# IP | Knapsack Problem

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Let  $v_j$  be the value of each item  $j$ , and  $W_0$  be the capacity of knapsack

- $x_j$  is 1 if item  $j$  is placed in knapsack, and 0 otherwise

$$\text{maximize} \quad \sum_{j=1}^n v_j x_j$$

$$\text{subject to} \quad \sum_{j=1}^n w_j x_j \leq W_0$$

$$x_j \in \{0, 1\}, \quad j = 1, 2, \dots, n$$

# IP | Knapsack Problem

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$$\begin{aligned} &\text{maximize} && \sum_{j=1}^n v_j x_j \\ &\text{subject to} && \sum_{j=1}^n w_j x_j \leq W_0 \\ &&& x_j \in \{0, 1\}, \quad j = 1, 2, \dots, n \end{aligned}$$



What if we have more than one knapsack?

# IP | Knapsack Problem

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What if we have more than one knapsack?

Then our notation changes and the problem becomes the **multiple knapsack problem!**

# IP | Multiple Knapsack Problem

---

Let  $m$  be the number of knapsacks, and  $W_i$  be the capacity of knapsack  $i$ .

- $x_{ij}$  is 1 if item  $j$  is placed in knapsack  $i$ , and 0 otherwise

$$\begin{aligned} &\text{Maximize} && \sum_{i=1}^m \sum_{j=1}^n v_j x_{ij} \\ &\text{subject to} && \sum_{j=1}^n w_j x_{ij} \leq W_i && \forall i = 1, 2, \dots, m \\ & && \sum_{i=1}^m x_{ij} \leq 1 && \forall j = 1, 2, \dots, n \\ & && x_{ij} \in \{0, 1\} && \forall i = 1, 2, \dots, m \quad \forall j = 1, 2, \dots, n \end{aligned}$$

# IP | Multiple Knapsack Problem

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$$\sum_{i=1}^m x_{ij} \leq 1 \quad \forall j = 1, 2, \dots, n$$

$$x_{ij} \in \{0, 1\} \quad \forall i = 1, 2, \dots, m \quad \forall j = 1, 2, \dots, n$$

? What if we have multiple compartments with varying capacities in each knapsack?



# IP | Multiple Knapsack Problem

Let  $m$  be the number of knapsacks, and  $W_i$  be the capacity of knapsack  $i$ .

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$$\text{subject to} \quad \sum_{j=1}^n w_j x_{ij} \leq W_i$$

$$\sum_{i=1}^m x_{ij} \leq 1$$

$$x_{ij} \in \{0, 1\}$$



What if we have multiple compartments with varying capacities in each knapsack?

$$\forall i = 1, 2, \dots, m$$

$$\forall j = 1, 2, \dots, n$$

$$\forall i = 1, 2, \dots, m \quad \forall j = 1, 2, \dots, n$$

Then our notation changes and the problem becomes a **multiple multi-dimensional knapsack problem!**

# IP | Multiple Multi Dimensional Knapsack Problem

Let  $m$  be the number of knapsacks, and  $d$  be the number of dimensions in each knapsack, and  $W_{ik}$  be the capacity of dimension  $k$  of knapsack  $i$ .

- $x_{ijk}$  is 1 if item  $j$  is placed in knapsack  $i$ 's dimension  $k$ , and 0 otherwise

$$\text{maximize} \quad \sum_{k=1}^d \sum_{i=1}^m \sum_{j=1}^n v_j x_{ijk}$$

$$\text{subject to} \quad \sum_{j=1}^n w_j x_{ijk} \leq W_{ik} \quad \forall i = 1, 2, \dots, m \quad \forall k = 1, 2, \dots, d$$

$$\sum_{k=1}^d \sum_{i=1}^m x_{ijk} \leq 1 \quad \forall j = 1, 2, \dots, n$$

$$x_{ijk} \in \{0, 1\} \quad \forall i = 1, 2, \dots, m \quad \forall j = 1, 2, \dots, n \quad \forall k = 1, 2, \dots, d$$

## Example | Multiple Knapsack Problem

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In the wake of a natural disaster, a humanitarian organization has gathered essential supplies to send to five affected regions. Each region has a limit on how much weight they can receive and distribute efficiently.

The organization must ensure that the supplies, each with its own weight and relief value, are distributed optimally.

# Example | Multiple Knapsack Problem

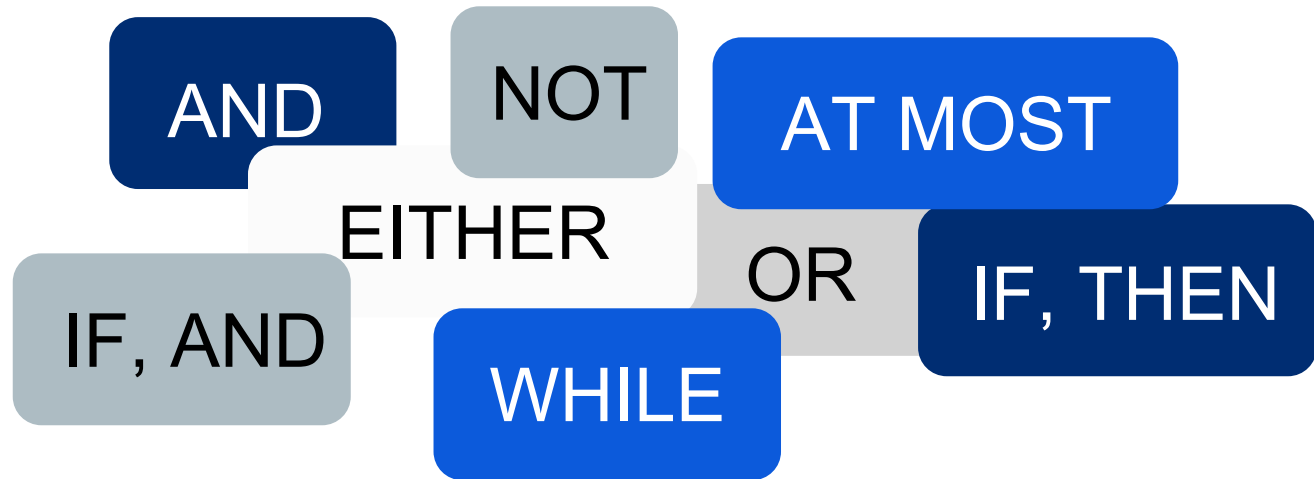
|        |               | Item Number | Supply              | Weight (kg) | Relief Value |
|--------|---------------|-------------|---------------------|-------------|--------------|
| Region | Capacity (kg) | 1           | Medical Kits        | 2           | 3            |
| A      | 5             | 2           | Fresh Water Packs   | 3           | 4            |
| B      | 8             | 3           | Tents               | 4           | 5            |
| C      | 7             | 4           | Warm Clothing       | 5           | 6            |
| D      | 6             | 5           | Non-perishable Food | 6           | 7            |
| E      | 4             | 6           | Hygiene Kits        | 1           | 0.8          |
|        |               | 7           | Blankets            | 4           | 3            |
|        |               | 8           | Lighting Supplies   | 5           | 4            |
|        |               | 9           | Communication Tools | 3           | 2            |
|        |               | 10          | Portable Stoves     | 2           | 3            |

# Modeling Logical Conditions

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One of the many powerful uses of binary variables is in the modeling of *logical* conditions

Some logical conditions:



Let's illustrate the logical features with a **capital budgeting** model

# Example | Capital Budgeting

Suppose a company is making allocation decisions on R&D investments, and there are 6 internal opportunities to consider:

| <b>Investment</b>             | <b>1</b> | <b>2</b> | <b>3</b> | <b>4</b> | <b>5</b> | <b>6</b> |
|-------------------------------|----------|----------|----------|----------|----------|----------|
| <b>Investment Cost (in K)</b> | \$5.50   | \$6.50   | \$7.00   | \$9.00   | \$10.50  | \$13.00  |
| <b>NetPresentValue (in K)</b> | \$15.00  | \$19.00  | \$19.00  | \$25.00  | \$29.00  | \$34.00  |

- Either an entire project investment is made, or not at all
- No investment can be made twice
- The present budget is \$23K and all unused portions of budget count toward total NPV

# Example | Capital Budgeting

Suppose a company is making allocation decisions on R&D investments, and there are 6 internal opportunities to consider:

| Investment             | 1       | 2       | 3       | 4       | 5       | 6       |
|------------------------|---------|---------|---------|---------|---------|---------|
| Investment Cost (in K) | \$5.50  | \$6.50  | \$7.00  | \$9.00  | \$10.50 | \$13.00 |
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- Either an entire project investment is made, or not at all
- No investment can be made twice
- The present budget is \$23K and all unused portions of budget count toward total NPV



What type of a problem is this?  
What's the optimal investment plan?

# Example | Capital Budgeting

---

For this problem, we consider:

- **all or nothing** capital budgeting decisions
- no investment can be made twice
- a budget cap exists on the maximum amount to invest
- each investment:
  - has a certain cost
  - returns a certain value



# Capital Budgeting Example

For this problem, we consider:

- **all or nothing** decisions,
- no investment can be made twice,
- a budget cap exists,
- each investment:
  - has a certain cost
  - returns a certain value

## Knapsack Problem

$$\text{Maximize } \sum_{j=1}^n v_j x_j$$

$$\text{subject to } \sum_{j=1}^n w_j x_j \leq w_0$$

$$x_j \in \{0, 1\}, \quad j = 1, 2, \dots, n$$

# Capital Budgeting Example

---

Then the IP becomes:

$$\text{Maximize } 9.5 x_1 + 12.5 x_2 + 12 x_3 + 16 x_4 + 18.5 x_5 + 21 x_6$$

$$\text{S.t. } 5.5 x_1 + 6.5 x_2 + 7 x_3 + 9 x_4 + 10.5 x_5 + 13 x_6 \leq 23$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \in \{0,1\}$$

# Capital Budgeting Example

---

Then the IP becomes:

$$\begin{aligned} \text{Maximize} \quad & 9.5 x_1 + 12.5 x_2 + 12 x_3 + 16 x_4 + 18.5 x_5 + 21 x_6 \\ \text{S.t.} \quad & 5.5 x_1 + 6.5 x_2 + 7 x_3 + 9 x_4 + 10.5 x_5 + 13 x_6 \leq 23 \\ & x_1, x_2, x_3, x_4, x_5, x_6 \in \{0,1\} \end{aligned}$$

While I am solving this IP using Gurobi, try it yourselves!

# Capital Budgeting Example

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What is the optimal investment strategy?

# Capital Budgeting Example

---

The optimal solution is to undertake **investments 1, 2, and 5**

- This uses \$22.5K of the \$23K available resources (i.e., within budget)
- The associated NPV with this decision is \$63K (\$15K + \$19K + \$29K)
- Plus \$.5K left over from unused budget, so total of **\$63.5K**

# Capital Budgeting Example

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- The associated NPV with this decision is \$63K (\$15K + \$19K + \$29K)
- Plus \$.5K left over from unused budget, so total of **\$63.5K**

! There is also another optimal solution!

- Undertake **investments 2, 3, and 4**

In Gurobi, we can find the other optimal solutions for an IP by taking an iterative approach – *might be time consuming for larger problems!*

# Capital Budgeting: Example & Logical Conditions

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Now suppose the company faces one of the following (mutually exclusive) events:

-  Invest in **either** investment 2 or 3, but **not both**
-  Investment in 4 **requires** investment in 6, and **vice versa**
-  At most **one** of investments 2, 4, **or** 5 can be invested in
-  Exactly **two** investments must be invested in
-  Investment in 2, **requires** investments in 4 **and** 6

# Choosing Between Investments

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0 stands for **False**

1 stands for **True**

When an **event takes place** we will represent it with **True** and when it is **not taking place** we will represent with **False**.

We were told that we can invest in only one Investment, 2 or 3, at once

So, **either** Investment 2 or Investment 3 can be True at the same time

| Investment 2 | Investment 3 |
|--------------|--------------|
| True (1)     | False (0)    |
| False (0)    | True (1)     |



# Event 1: Can do 2 or 3, *but not both*

---

One tool to assist with modeling logical conditions is a ***truth table***

- List out all possible combinations of binary variable values:

| Investment 2 | Investment 3 |
|--------------|--------------|
| 0            | 0            |
| 0            | 1            |
| 1            | 0            |
| 1            | 1            |

- What's the equivalent constraint for this event?

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| 1            | 0            |
| 1            | 1            |

- What's the equivalent constraint for this event?

$$x_2 + x_3 \leq 1$$

Because there are two optimal solutions to the problem, the solution to undertake 1, 2, and 5 is still possible, and so this condition **does not impact** the optimal objective function value

## Event 2: If 4, then 6... and vice versa

---

List out all possible combinations of binary variable values:

| Investment 4 | Investment 6 |
|--------------|--------------|
| 0            | 0            |
| 0            | 1            |
| 1            | 0            |
| 1            | 1            |

○ What's the corresponding constraint for this event?

## Event 2: If 4, then 6... and vice versa

List out all possible combinations of binary variable values:

| Investment 4 | Investment 6 |
|--------------|--------------|
| 0            | 0            |
| 0            | 1            |
| 1            | 0            |
| 1            | 1            |

○ What's the corresponding constraint for this event?

$$x_4 - x_6 = 0$$

The original optimal solution of undertaking 1, 2, and 5 is still optimal, but the alternate optimal solution of undertaking 2, 3, and 4 will not work.

## Event 3: At most one of 2, 4, and 5

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List out all possible combinations of binary variable values:

What's the corresponding constraint for this event?

| Investment 2 | Investment 4 | Investment 5 |
|--------------|--------------|--------------|
| 0            | 0            | 0            |
| 0            | 0            | 1            |
| 0            | 1            | 0            |
| 0            | 1            | 1            |
| 1            | 0            | 0            |
| 1            | 0            | 1            |
| 1            | 1            | 0            |
| 1            | 1            | 1            |

## Event 3: At most one of 2, 4, and 5

List out all possible combinations of binary variable values:

What's the corresponding constraint for this event?

$$x_2 + x_4 + x_5 \leq 1$$

| Investment 2 | Investment 4 | Investment 5 |
|--------------|--------------|--------------|
| 0            | 0            | 0            |
| 0            | 0            | 1            |
| 0            | 1            | 0            |
| 0            | 1            | 1            |
| 1            | 0            | 0            |
| 1            | 0            | 1            |
| 1            | 1            | 0            |
| 1            | 1            | 1            |

- No longer will any of the original optimal solutions work
- The new optimal solution is to undertake investments 1, 3, and 5
  - This leads to a new optimal objective function value of \$63K

## Event 4: Exactly 2 of 6 Investments

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This one may be more obvious than previous ones...

- What's the corresponding constraint?

## Event 4: Exactly 2 of 6 Investments

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This one may be more obvious than previous ones...

- What's the corresponding constraint?

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 2$$

The new optimal solution is to undertake investments 4 and 6

- This leads to a new optimal objective function value of (only) \$60K



## Event 5: If 2, then 4 and 6

List out all possible combinations of binary variable values:

What's the corresponding constraint for this event?

| Investment 2 | Investment 4 | Investment 6 |
|--------------|--------------|--------------|
| 0            | 0            | 0            |
| 0            | 0            | 1            |
| 0            | 1            | 0            |
| 0            | 1            | 1            |
| 1            | 0            | 0            |
| 1            | 0            | 1            |
| 1            | 1            | 0            |
| 1            | 1            | 1            |

## Event 5: If 2, then 4 and 6

List out all possible combinations of binary variable values:

What's the corresponding constraint for this event?

| Investment 2 | Investment 4 | Investment 6 |
|--------------|--------------|--------------|
| 0            | 0            | 0            |
| 0            | 0            | 1            |
| 0            | 1            | 0            |
| 0            | 1            | 1            |
| 1            | 0            | 0            |
| 1            | 0            | 1            |
| 1            | 1            | 0            |
| 1            | 1            | 1            |

There are actually two ways to model this!

1. Model with two constraints:

$$x_4 - x_2 \geq 0 \text{ and } x_6 - x_2 \geq 0$$

2. A single constraint:

$$0.5x_4 + 0.5x_6 - x_2 \geq 0$$

## Event 5: If 2, then 4 and 6

List out all possible combinations of binary variable values:

What's the corresponding constraint for this event?

| Investment 2 | Investment 4 | Investment 6 |
|--------------|--------------|--------------|
| 0            | 0            | 0            |
| 0            | 0            | 1            |
| 0            | 1            | 0            |
| 0            | 1            | 1            |
| 1            | 0            | 0            |
| 1            | 0            | 1            |
| 1            | 1            | 0            |
| 1            | 1            | 1            |

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Which is better?

- It turns out that the 1<sup>st</sup> is *stronger* than the 2<sup>nd</sup>
  - Provides more information
- Can derive second from the 1<sup>st</sup>, *but not first from 2<sup>nd</sup>*

## Integer Programming Examples (Add as a Quiz Question Later)

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**Question:** If we want to restrict  $x$  to be one of the elements  $\{2, 7, 12\}$ , how should we model it?

$$\begin{aligned}x &= 2w_1 + 7w_2 + 12w_3 \\w_1 + w_2 + w_3 &= 1 \\w_i &\in \{0,1\} \text{ for } i = 1 \text{ to } 3\end{aligned}$$

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$$\begin{aligned}x &= 2w_1 + 7w_2 + 12w_3 \\w_1 + w_2 + w_3 &\leq 1 \\w_i &\in \{0,1\} \text{ for } i = 1 \text{ to } 3\end{aligned}$$

# Integer Programming Examples

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**Question:** How can we restrict  $x$  to be either 0 or between particular positive bounds?

**In algebraic notation:**  $x = 0 \text{ or } l \leq x \leq u$

**Define:** 
$$y = \begin{cases} 0, & \text{for } x = 0 \\ 1, & \text{for } l \leq x \leq u \end{cases}$$

**Then, the logical constraints can be modeled by:**

$$ly \leq x \leq uy, \quad \text{where } y \in \{0,1\}$$

# Integer Programming Examples

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**Question:** If we want to have either one of these constraints:

$2x_1 + 3x_2 \geq 14$  **or**  $5x_2 - 7x_3 \leq 3$  how should we model it?

$$2x_1 + 3x_2 \geq 14 - M(1 - w)$$

$$5x_2 - 7x_3 \leq 3 + Mw$$

$$w \in \{0,1\}$$